MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level
Higher 2

MATHEMATICS

9758/01

Paper 1

October/November 2025

3 hours

Additional Materials: Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.
Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

The number of marks is given in brackets [ ] at the end of each question or part question.

1 A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_1 = a,$$

$$u_{r+1} = 3u_r + 5, \quad r \geqslant 1,$$

where a is a constant.

(a) Find expressions for u_2 and u_3 in terms of a. [2]

(b) It is given that $\sum_{r=1}^{4} u_r = 110$. Find the value of a. [3]

2 **(a)** The sum of the first 20 terms of an arithmetic sequence is 65. The sum of the next 8 terms is -30.

Find the first term and the common difference of this sequence. [3]

(b) A sequence has nth term given by $u_n = \mathrm{e}^{k(n+1)}$, where k is a constant and $k \neq 0$.

Show that this sequence is geometric. [2]

3 **Do not use a calculator in answering this question.**

A complex number z is given by $z = \dfrac{\sqrt{3}+\mathrm{i}}{\sqrt{3}-\mathrm{i}}$.

(a) Find $|z|$ and $\arg(z)$. [4]

The point A represents the complex number z on an Argand diagram. The point A is rotated anticlockwise through $\frac{1}{2}\pi$ about the origin O to the point B.

(b) Plot the points A and B on an Argand diagram. State in terms of z the complex number that is represented by B, explaining your answer. [3]

4 A curve is given by the equation $y = \dfrac{3x+10}{\sqrt{x+1}}$.

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ and hence find the x-coordinate of the turning point of the curve. [3]

(b) Use calculus to find the value of the second derivative at the turning point and state the nature of this turning point. [3]

5 **(a)** Solve exactly the inequality $\mathrm{e}^x \geqslant 2\mathrm{e}^{-x} + 1$. [3]

(b) Hence find $\displaystyle\int_0^1 \left|\mathrm{e}^x - 2\mathrm{e}^{-x} - 1\right| \mathrm{d}x$, giving your answer in exact form. [4]

6 The carbon content of plants includes a small proportion of the radioactive isotope carbon-14. The mass of carbon-14 in dead plants decreases over time. Scientists measure the mass of carbon-14 in plants to estimate how long ago they died.

The mass of carbon-14 in dead plants decreases by 50% in 5730 years.

The mass, M, of carbon-14 in a plant t years after it has died is modelled by the differential equation

$$\frac{dM}{dt} = -kM,$$

where k is a positive constant. The initial value of M is M_0.

(a) By solving the differential equation, find the value of k. [4]

(b) A dead plant has 20% of its original mass of carbon-14 remaining. Determine, to the nearest 100 years, how long ago the plant died. [2]

7 **(a)** Show that $\frac{d}{dx}(x\ln x - x) = \ln x$. [2]

A curve C is given by the equation $y = \ln x$. The region R is bounded by C, the x-axis and the line $x = e$.

(b) Using the result in part **(a)**, find the exact volume generated when R is rotated 360° about the x-axis. [4]

(c) Find the volume generated when R is rotated 360° about the y-axis, giving your answer correct to 2 decimal places. [3]

8 The plane π_1 has equation $2x + y + 3z = 2$. A point P has position vector $\begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}$.

(a) Find the foot of the perpendicular from P to the plane π_1. [4]

(b) Find the position vector of the reflection of P in the plane π_1. [2]

A second plane π_2 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + \mu\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$, where λ and μ are parameters.

(c) Find the acute angle between π_1 and π_2. [3]

9 It is given that $y = \sin^{-1} px$, where p is a constant.

(a) Show that $\dfrac{dy}{dx} = p \sec y$. [2]

(b) By further differentiation of the result in part **(a)** find, in terms of p, the Maclaurin series for y up to and including the term in x^3. [5]

(c) Use your answer to part **(b)** to find the series expansion of $\dfrac{1}{\sqrt{1-9x^2}}$, up to and including the term in x^2. [3]

10 The function f is given by

$$\mathrm{f}: x \mapsto \frac{4-ax}{x+b}, \qquad x \in \mathbb{R}, \qquad x \neq -b,$$

where a and b are positive constants.

(a) Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes and the coordinates of any points where the curve crosses the axes. [4]

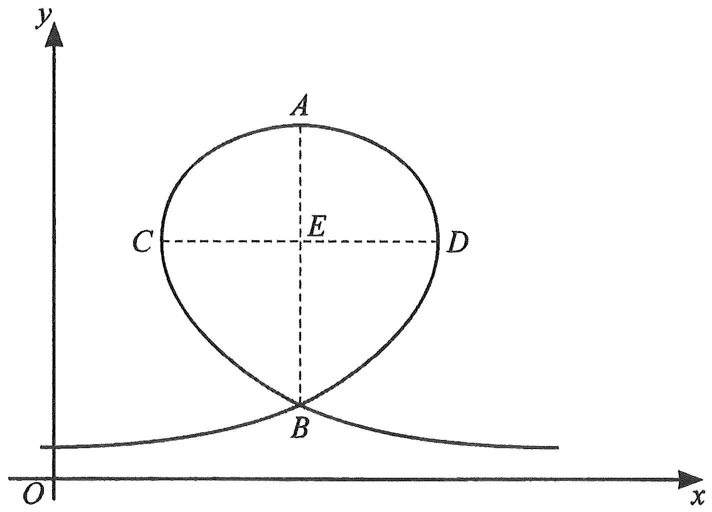
(b) Describe a sequence of 3 transformations that would transform the graph of $y = \mathrm{f}(x)$ onto the graph of $y = \dfrac{1}{x}$. [4]

(c) Find $\mathrm{f}^{-1}(x)$ and state its range. [3]

It is now given that $b = a$.

(d) Find the value of the composite function $\mathrm{f}^{2025}(x)$ when $x = 2$, giving your answer in terms of a. [1]

11



Engineers are designing a roller coaster. The diagram shows the side view of a section of track with a loop. This section is modelled by the curve with parametric equations

$$x = 28(\theta\cos\theta + 1), \qquad y = 20(2 - \theta\sin\theta),$$

for $-\frac{2}{3}\pi \leqslant \theta \leqslant \frac{2}{3}\pi$, where x and y are measured in metres. The x-axis models the horizontal ground.

The maximum point on the curve is at A, and the line AB is vertical. Points C and D are where the tangent to the curve is parallel to the y-axis. Point E is the intersection of the lines AB and CD.

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ. [3]

(b) **(i)** Find the values of θ at points C and D, giving your answers correct to 2 significant figures. [3]

(ii) Hence find the greatest width, CD, of the loop. [2]

The x-coordinate of point A is 28.

(c) **(i)** Find the value of θ at point A. [1]

(ii) Find the 2 possible values of θ at point B. [2]

(iii) Hence find the distance AB. [1]

Engineers know that, for safety reasons, the distance AE should be within 10% of the distance $\frac{1}{2}CD$.

(d) Determine whether the design of this section of track satisfies this condition. [2]

Question 12 is printed on the next page.

12 **Do not use a calculator in answering this question.**

It is given that $1+2\mathrm{i}$ is a complex root of the equation $z^3+pz^2+qz-10=0$, where p and q are real numbers.

(a) State another complex root of the equation in the form $a+\mathrm{i}b$, where a and b are real numbers and $b \neq 0$. Justify your answer. [2]

(b) Find the values of p and q, and find the third root. [6]

(c) Hence find the roots of the equation $1+pw+qw^2-10w^3=0$. [2]